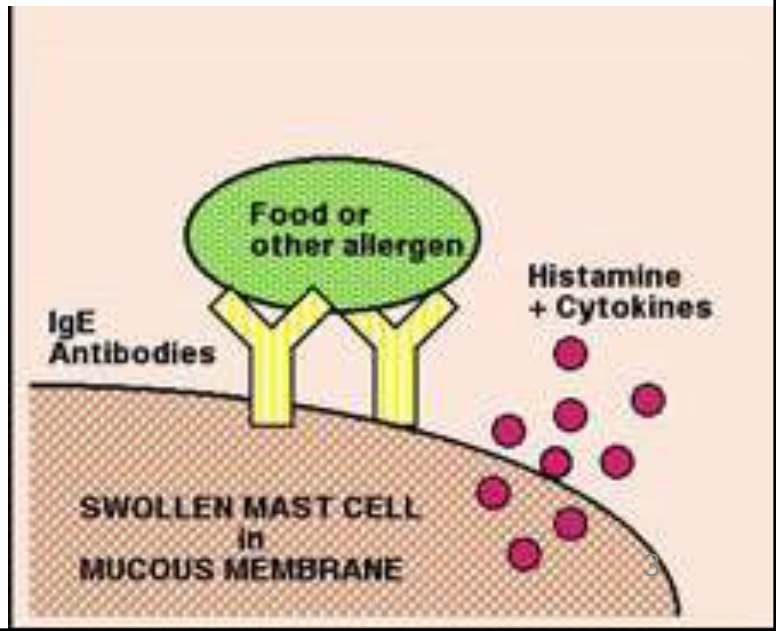
Allergic Lid Disease

Lorne Yudcovitch, O.D., M.S., F.A.A.O.

ALLERGIC DISEASE BACKGROUND

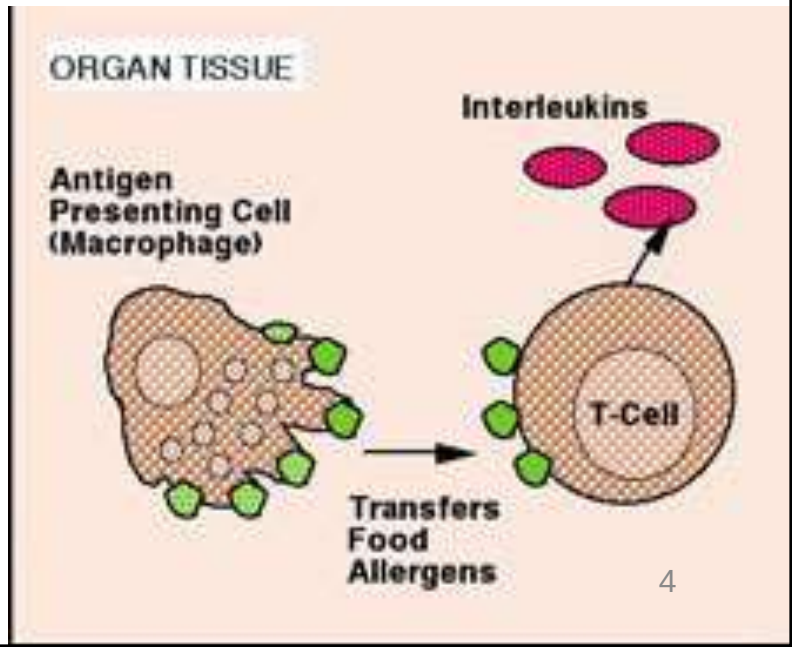
Types of allergic reactions

- Type I - immediate type reaction:
- Reaction resulting from the release of pharmacologically active substances such as histamine, or eosinophillic chemotactic factor (ecf) from IgE-sensitized basophils and mast cells after contact with specific antigens
- Occurs within minutes to hours



Types of allergic reactions

- Type IV - delayed type reaction:
- Reactions caused by sensitized lymphocytes (T-cells) after contact with antigens which result from direct cytotoxicity or from the release of lymphokines
- Cell-mediated reaction
- Occurs over hours to days



A key allergic clinical finding…

- **Dennie–Morgan fold (aka Dennie-Morgan line)**
- A fold or line in the skin below the lower eyelid caused by edema in allergic dermatitis

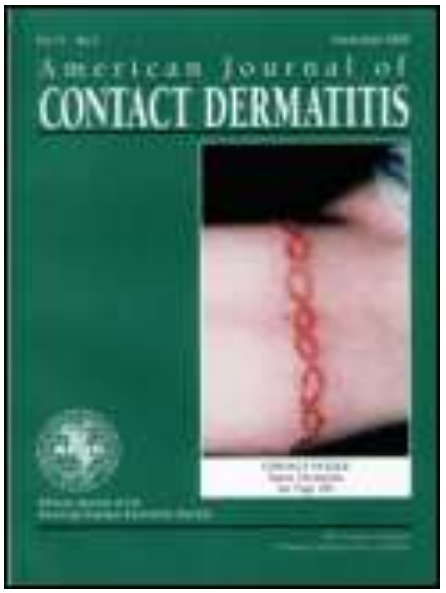




CONTACT DERMATITIS

Contact Dermatitis







CONTACT DERMATITIS



- Very common immunological disease
- Results from direct contact with environmental allergen
- Dermatitis = any inflammation of the epidermis
- Contact dermatitis = any skin inflammation that occurs when the skin surface comes in contact with a substance originating outside the body
- Episodes can be triggered by specific events

Irritant Contact Dermatitis



- More common
- Contact with substance that directly damages skin
- Nonspecific response of skin to chemical damage that releases mediators of inflammation from epidermal cells
- Occurs within 1 – 24 hours; can occur within minutes

Irritant Contact Dermatitis

- Hx is very important!
- Red rash with pain, burning, stinging, discomfort; maybe no itching
- Often affects hands
- Occupations: cleaning, hospital care, food prep, hairdressing
- Women > men
- Common irritants
- Soap and water, solvents, microtrauma, dry air, cleaners



Irritant Contact Dermatitis









Irritant Contact Dermatitis

- Erythema
- Edema
- Glazed, parched, scalded appearance
- Skin usually dry with scaling
- No diagnostic test
  - Patch test negative
- Dx based on exclusion of other cutaneous diseases and clinical appearance of site exposed to a known irritant



Irritant Contact Dermatitis





Irritant Contact Dermatitis

TX/Management



- Avoid irritant via barriers
- Non-sensitizing moisturizers
  - Plain petroleum jelly
- Mild skin cleansers
  - Aquanil, Cetaphil cleanser, Neutrogena cleanser



- Soaps with low irritant effect
  - Dove, Cetaphil
- Alcohol based hand cleansers with moisturizer
- Avoid inappropriate solvents or abrasives to clean hands
- No topical steroids!!!

ICD Management









Allergic Contact Dermatitis:

- Occurs in sensitized individuals
- Involves a cell mediated immunity response
- Requires a latency period
- The re-exposure precipitates the dermatitis
  - Response is usually in 48 hrs

Subjective

- Hx is very important!
- Red rash with ITCHING



Allergic Contact Dermatitis:

List of possible allergens is endless

- Plants (poison oak*, ivy)
- Metals (nickel*, gold)
- Hair products
- Skin care products
- Rubber/latex
- Dyes
- Chromates for leather tanning
- Fragrances
- Preservatives
- Acrylates
- Cosmetics







Allergic Contact Dermatitis to Nickel





Allergic Contact Dermatitis to Hair Spray



Allergic Contact Dermatitis to Cosmetics



Examples of medications that may *cause* Allergic Contact Dermatitis

Topical ocular drugs

- Antazoline
- Neomycin
- Phenylephrine
- Atropine
- Pilocarpine
- Anti-glaucoma drugs





Allergic Contact Dermatitis due to Topical Ocular Drug





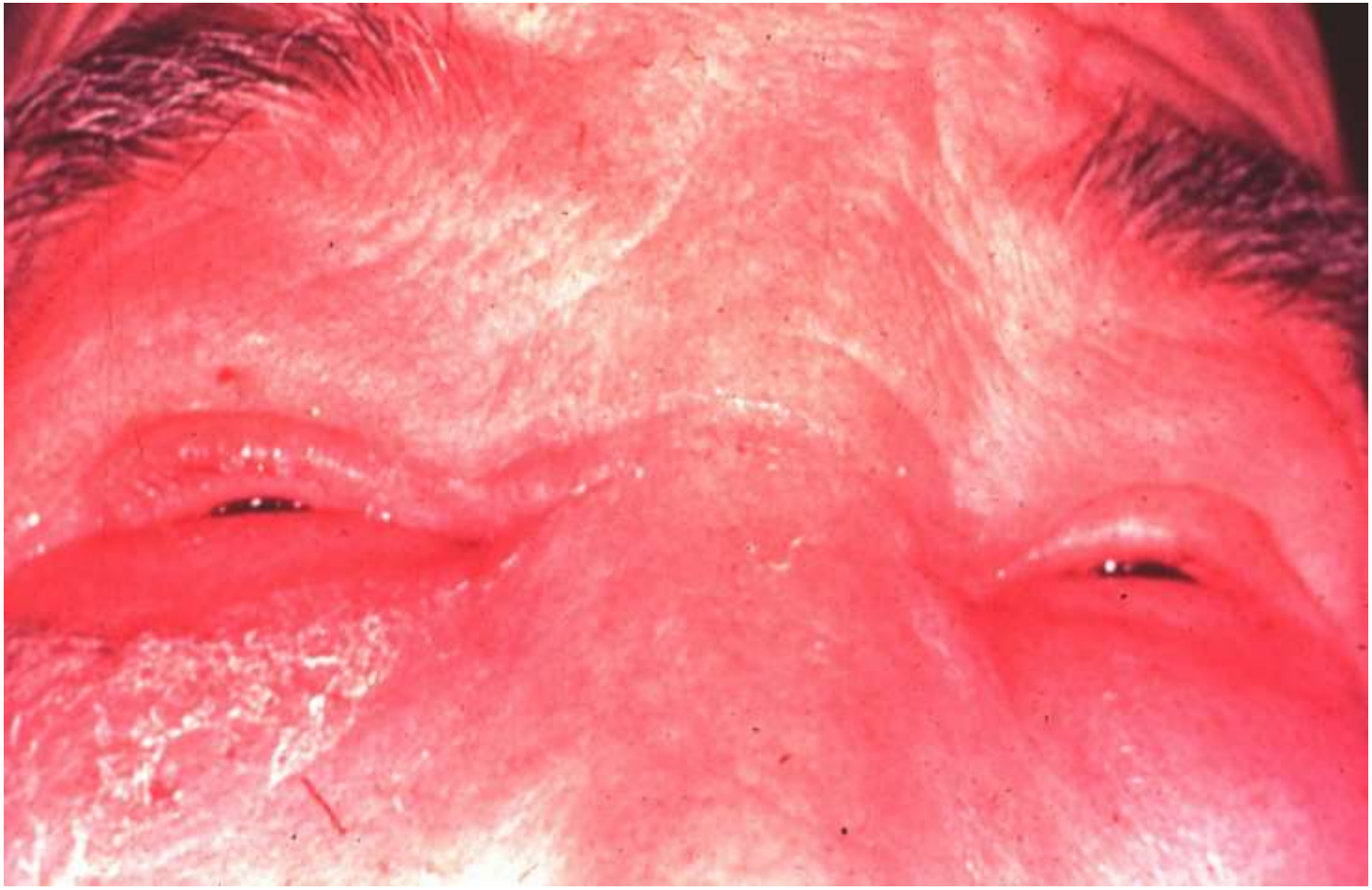
ACD due to Topical Neomycin



ACD to Topical Ocular Atropine



ACD to Topical Ocular Mydriatic



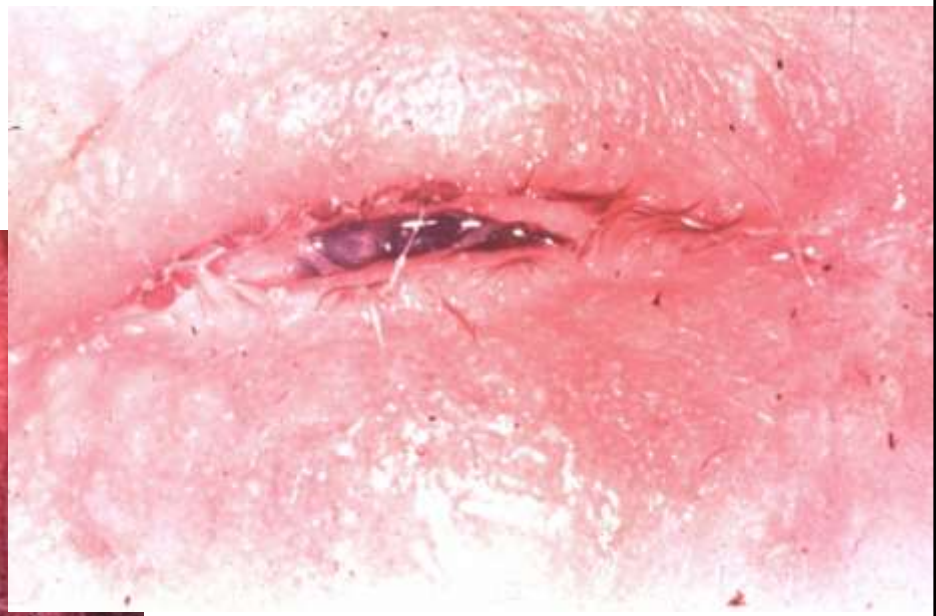
Allergic Contact Dermatitis

Objective

- Pruritis (= itch!)
- Erythema
- Edema
- Chemosis of conjunctiva
- Vesicles
- Weeping rash → crusting, scaling → darkened, cracked (fissured), leathery appearance
- Keratitis may develop



Allergic Contact Dermatitis to Contact Lens Solution





Conjunctival Chemosis



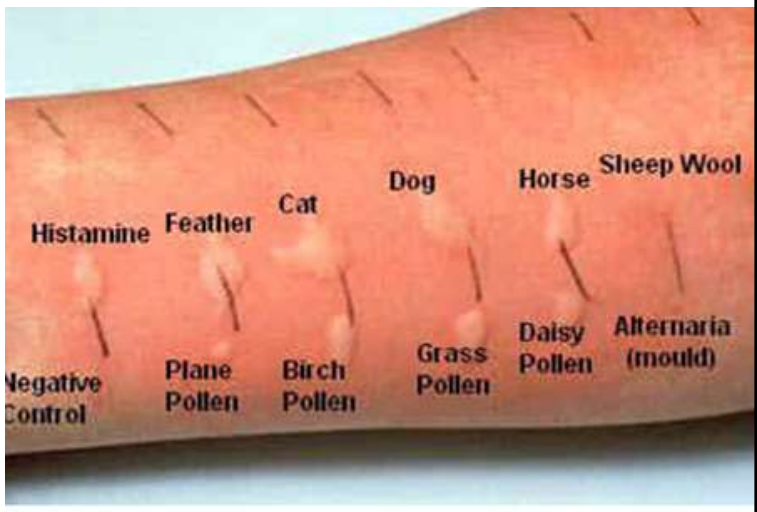
Conjunctival Chemosis



Allergic Contact Dermatitis

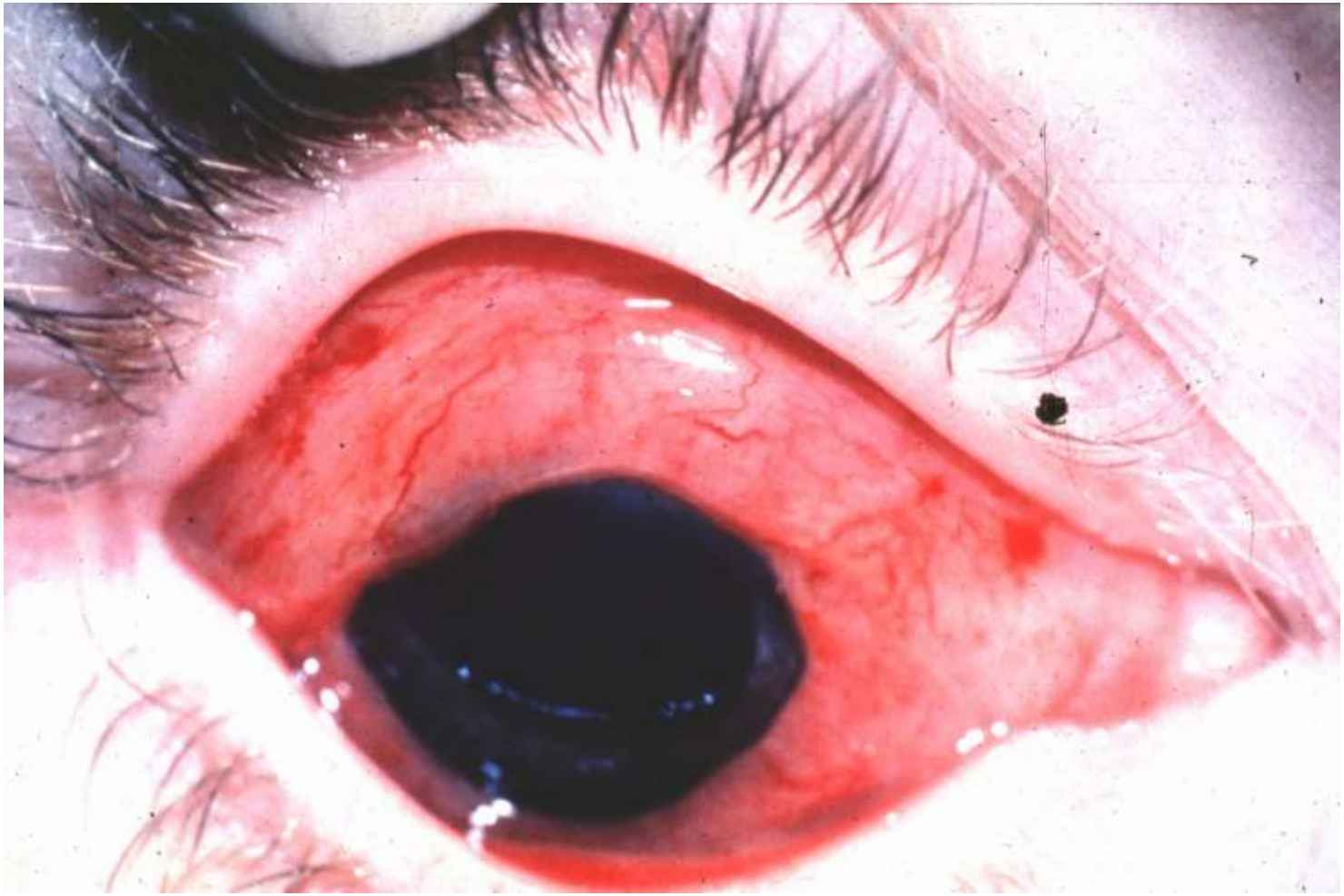
DX

- Elicited from history
- Patch skin tests
  - Application of suspected materials to skin on patches
  - Left on 48 hours, then read again between 72 hrs – 1 wk
  - Positive: development of erythema, papules, edema, vesicles
  - Correlation with history, location of eruption, clinical course to determine if true etiology
- DDX from many similar appearing rashes



Skin Allergy Test

Conjunctival Chemosis



ACD





Allergic Contact Dermatitis

TX/Management

- Remove allergen
- Cold compresses 4 – 6X/day
- Topical steroid ung or cream BID – TID for 5 – 10 days
  - < 1% hydrocortisone
- Oral antihistamine
  - Diphenhydramine (Benadryl)
- Ocular lubricants 4 – 6X/day for comfort or keratitis
- Topical immunomodulator BID
  - Pimecromlimus (Elidel cream)
  - Tacrolimus (Protopic)





ACD Management







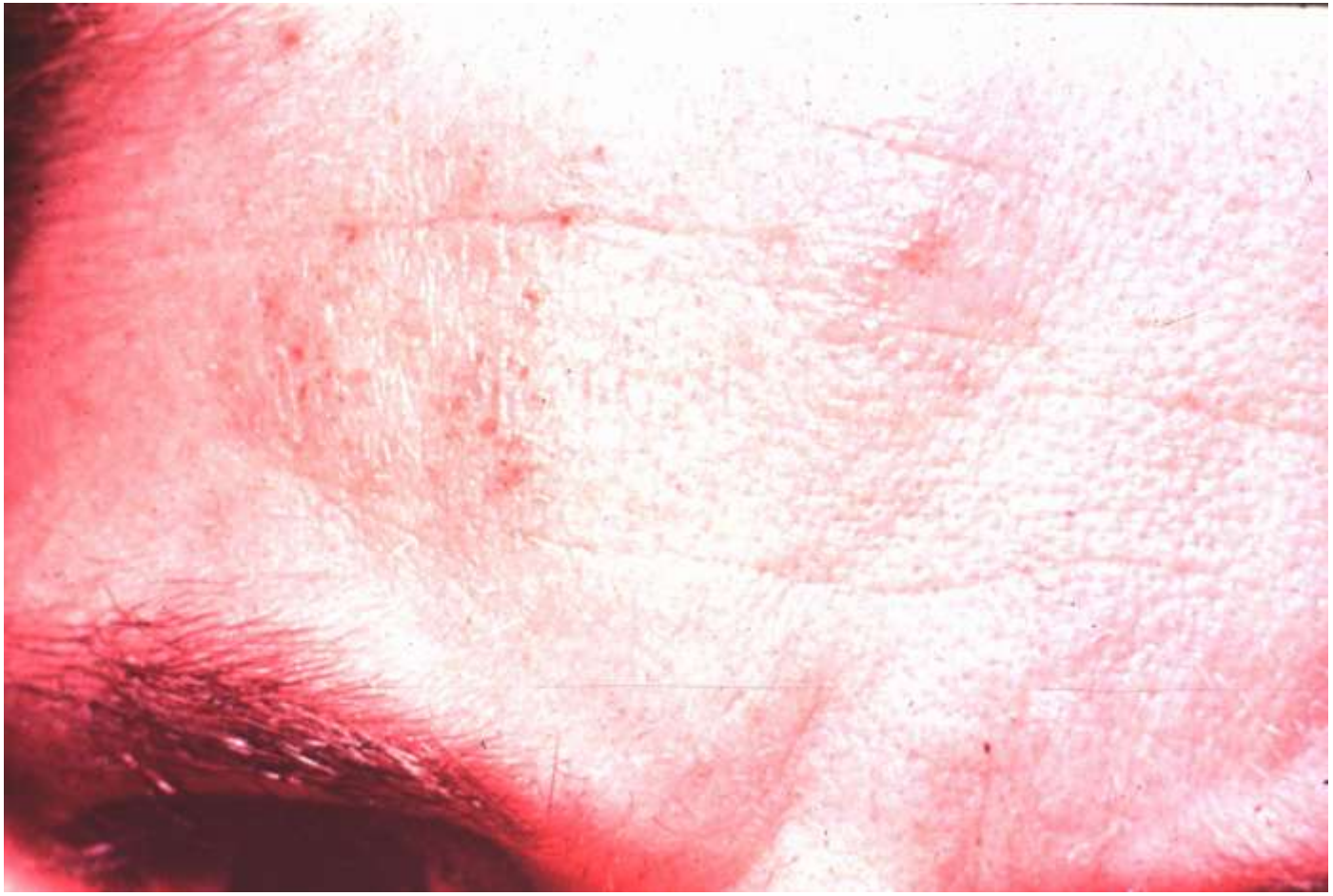


ATOPIC DERMATITIS

- Eczematous skin eruption AKA eczema
- Onset in infancy: most common skin disease in children
- Inherited condition
  - Usually in conjunction with other atopic conditions
- INFLAMED PATCHES OF VERY ITCHY SKIN



Atopic Dermatitis



Atopic Dermatitis

Clinical manifestations

- Lesions vary with age



Early infancy:

- Rash that is exudative, vesicular to maculopapular, see fine scales
- Location often on face (cheeks common), scalp, knees in infants
- ITCHING

Atopic Dermatitis



Atopic Dermatitis





Atopic Dermatitis

Later stages (as pt. ages):



- Dermatitis
- Skin becomes dry and lichenified
- Excoriated areas (abraded)
- Pruritus→ ITCHING
- Healing - hypo or hyper-pigmentation
- Location most often seen on elbows, neck, back of knees, ankles in adults

Atopic Dermatitis



Atopic Dermatitis
Eczematous flakes



Atopic Dermatitis



Atopic Dermatitis





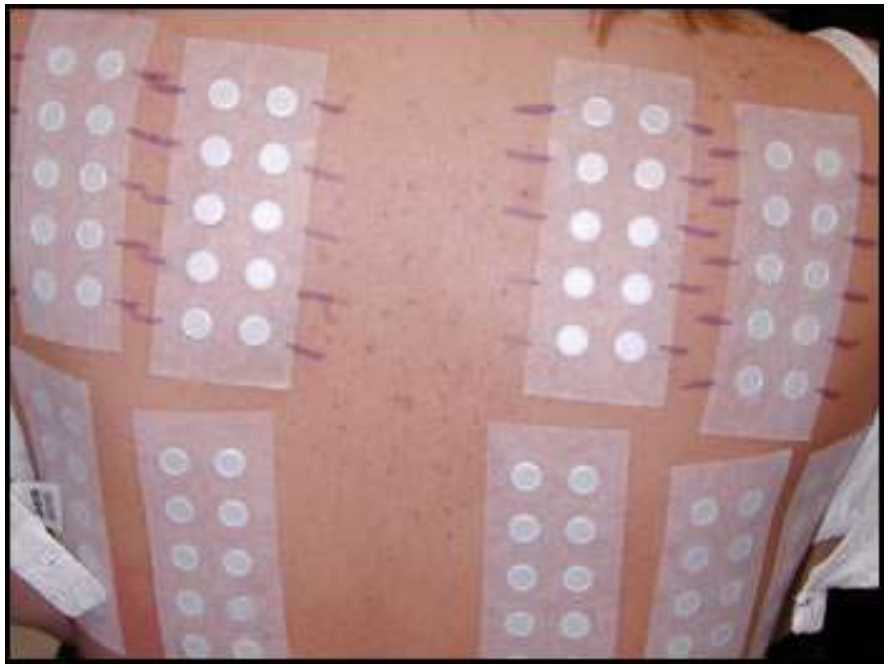


Atopic Dermatitis

- Patient is more prone to secondary infection due to Staphylococcus or Streptococcus
- Unusual susceptibility to vaccinia, herpes simplex and molluscum contagiosum viruses
- Possible ocular manifestations of atopic dermatitis
  - Primarily lids
- Clinical Course unpredictable, but usually subsides by age 3 or 4 with exacerbations and remissions frequently recurring throughout childhood, adolescence and adulthood

Atopic Dermatitis

- ITCHY rash
- Erythema
- Edema
- Exudative lesions
- Xerosis
- Lichenfication
- Excoriation and crusting
- DX - tests not needed if condition presents with clear clinical markers
  - Patch test



Patch Test





Atopic Dermatitis TX/Management





- Cool damp compresses x 15-30 min or short, cool shower/bath
- White petrolatum moisturizer
- Topical steroid
  - 1% hydrocortisone ung BID – TID
- Oral antihistamine
  - Diphenhydramine (Benadryl)
- Topical antibiotic if lesions secondarily infected
- Topical immunomodulator
  - Pimecromlimus (Elidel cream)
  - Tacrolimus (Protopic)
- Nonmedical considerations:
  - Clothing
  - Cool temperature
  - Humidifier



AD Management













URTICARIA AND ANGIOEDEMA (Hives and Giant Urticaria)

Urticaria

- Urticaria = ITCHY WELTS = HIVES:
- Cutaneous eruption
- Multiple pathogenic mechanisms
- May or may not be immunologic in origin
- Only 70% have a specific identifiable cause
- Acute urticaria -- immunologic
- Chronic urticaria -- unknown



Urticaria

Urticaria



Urticaria

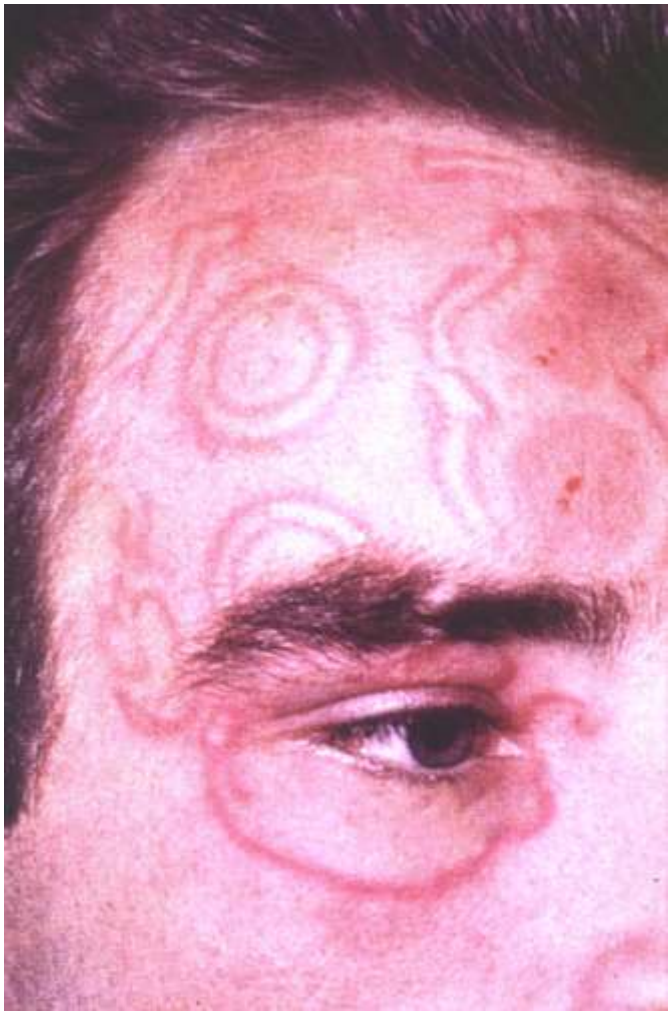
- 10 to 25% of the population
- Intense pruritus
- Raised wheals surrounded by erythematous flare
  - Size 1 to 5 mm to greater than 20 mm
  - Dilation and increased permeability of blood vessels in superficial dermis with edema in surrounding epidermis
  - May have a ring shape
- May occur on any part of the body
- May be transient in location

Urticaria





Dermatographic Urticaria
aka Dermatographism aka “skin writing”





ANGIOEDEMA: WHEN THE INFLAMMATION INVOLVES DEEPER LAYERS OF SKIN

- Skin appears normal
- Dilation and increased permeability of blood vessels in the deep dermis with resulting edema of dermis
- More extensive edematous areas
- Little pruritus - sometimes described as painful or burning
- Most often involves face, lips, eyelid, tongue, throat, hands, feet
- May produce respiratory distress due to upper airway edema



Angioedema



Angioedema



URTICARIA AND ANGIOEDEMA CAUSES

- Drug reactions
- Foods, food additives
- Inhalant, ingested or contact antigens
- Infections
- Physical
  - Cold
  - Heat
  - Exercise – induced
  - Solar
  - Water
- Insect bites
- Psychogenic
- Systemic disease





Angioedema



URTICARIA AND ANGIOEDEMA

DDX:

- Need to be concerned that the edema is not caused by renal or cardiac disease
- Contact dermatitis



TX:

- Try to find the cause and avoid
- Acute phase:
  - Cold compresses
  - Antihistamine
  - Systemic steroids - prednisone unpredictable
  - Epinephrine sc for acute pharyngeal or laryngeal angioedema

THANK YOU!